SETTLE The SETTLE method [274] is just the SHAKE approach applied to a small system combined with an exact solve of the nonlinear equations. It is usually applied to treat rigid bodies (as discussed in the following section).

LINCS In solving the Eqs. (4.27)–(4.29) by Newton iteration, the largest cost and the most significant barrier to parallelization is the linear solve. The LINCS method [119, 175] uses a series expansion to approximate the inverse matrix (which is valid only under certain conditions). The method gives substantial speedups in large, weakly-coupled systems.

It is also possible to use internal coordinate representations to treat constraints, i.e. parameterization schemes, which are typically based on partitioning the coordinates of the system into a set of independent variables and a complementary collection of dependent variables (which are expressed in terms of the first set). These approaches result in systems of slightly smaller dimension, but typically greater complexity. Moreover, the parameterization is typically local which leads to additional implementation issues. For some details on such schemes and discussion of their use in molecular simulation, the reader is referred to [20, 190, 255].

4.3.7 Numerical Experiment with SHAKE

We demonstrate the numerical stability benefit of using constraints in simulation by considering a periodic box of 116 TIP3P water molecules in a periodic box with side length 16A (for computational convenience the simulated particle density is slightly below that of liquid water). We simulate the system using the NAMD package [302], with the CHARMM22 forcefield [246] parameterizing the potential energy functions. After an initial equilibration of 1ns, we look at the stability and conservation of energy in this system, when using either the constrained or unconstrained version of the model. The deviation in the total energy over a further 1ns simulation is plotted in Fig. 4.5. In the unconstrained version of the model, the internal angle and the bond lengths of each molecule are free to change, governed by a restraining force from the corresponding potential energy terms. This model proves to be unstable for stepsizes chosen at 3 fs and over, while the simulation at $h = 2$ fs showed a large variance in the energy error (as well as exhibiting energy drift in longer simulations). For accurate results, we would expect the stepsize to be restricted to around $h = 1$ fs, where there is significantly smaller average energy fluctuation.

By contrast, when using constraints to keep the angle and bond lengths fixed, we observe that the energy fluctuations are dramatically reduced and we are able to use considerably higher stepsizes than in the unconstrained case. Although the use of constraints slightly increases the required computational workload (relative to the flexible model), the non-bonded force calculation (especially the Coulombic component) is typically still the dominant cost in each step of the simulation. As the SHAKE method does not require extra force evaluations, the ability to increase